

### SECTION-D

**Note:** Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain Hybrid Electric Vehicles (HEV) with their configurations.
- Q.24 Explain the all cause of battery explosions.
- Q.25 Define the PHEV (Plug-in Hybrid Electric Vehicle) and its components with diagram.

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### 4th Sem / Instrumentation & Control

### Subject : Electric Vehicles

Time : 3 Hrs.

M.M. : 60

### SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Which factor affects the lifespan of a lithium-ion battery in Evs?
- a) Number of charging cycles
  - b) Temperature
  - c) Depth of discharge
  - d) All of the above
- Q.2 Traction inverter is a power electronic device that is used in electric and hybrid vehicles to convert the \_\_\_\_\_
- a) DC to AC
  - b) AC to DC
  - c) DC to DC
  - d) AC to AC
- Q.3 Which Battery are preferred for EV
- a) Lead-Acid (Pb - acid)

- b) Lithium-Ion (Li-ion)
- c) Sodium-Sulphur (NaS)
- d) Nickel-Cadmium (NiCd)

Q.4 Fuel Cell use combination of

- a) Zinc + Sulphur      b) Sulphur + Oxygen
- c) Hydrogen + Oxygen   d) Sodium + Sulphur

Q.5 Battery that cannot be charged again is called

- a) Primary Battery
- b) Secondary Battery
- c) Both Primary and Secondary
- d) Neither Primary Nor secondary

Q.6 What does kWh stand for in EV battery specifications?

- a) Kilowatt                      b) Kinetic Watt hour
- c) Kilowatt-hour              d) Kelvin Watt

### SECTION-B

**Note:** Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 What is the unit of charge capacity in a battery.

Q.8 \_\_\_\_\_ Vehicles are powered by battery only

Q.9 What do you mean by the open circuit in a battery?

Q.10 Write the full form of SRM.

Q.11 Battery that can be charged again is called \_\_\_\_\_?

Q.12 Battery voltage is same in parallel (T/F)

### SECTION-C

**Note:** Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 What are the Social & Environmental importance of Hybrid and Electric Vehicles?

Q.14 Explain energy measurement system in short.

Q.15 What do you mean by the Super-capacitors used in EV.

Q.16 Write the working of electrochemical cell & charges on anode and cathode.

Q.17 What are the basic component of Fuel Cell Electric Vehicle.

Q.18 Define the International Standards of DC charging

Q.19 What is state of charge (SOC) in battery?

Q.20 Write the short notes on Inductive charging.

Q.21 What are the basics of Micro, Mild, and Mini EV.

Q.22 Write a short note on the environmental benefits of using EVs.